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Description - This calibration compensates for errors in table positioning. Errors can be caused by differing encoders and stacking tolerances in the drive system components.

This calibration is normally required after initial start-up of software on a new system. An error is logged to indicate the need for this calibration (see Screen 1).

SCREEN 1

EM_SRI_INVALID_CALIB_VALUE

The encoder calibration value was invalid.

Please run the cradle calibration routine.

Specified Calibration Constant: (Difference between current **endTravel**
and **begTravel**)

Minimum Calibration Constant: 2250

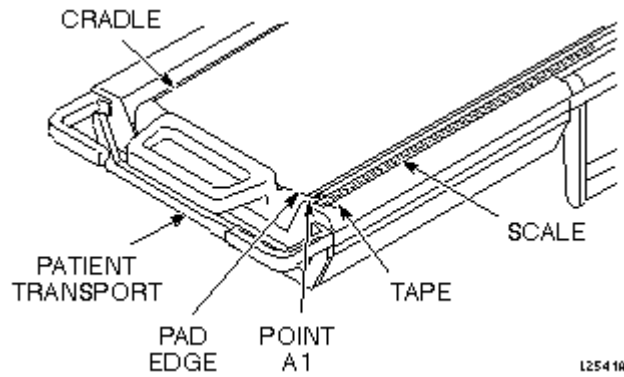
Maximum Calibration Constant: 2750

This error is only a warning, and indicates that the table calibration files are not correct. Calibration may be necessary if the image does not align with the landmark. However, other factors could be causing this error, and calibration may not be the solution.

An alternative proprietary procedure is available for GE use and to sites with a valid Advanced Service Package Limited License. Refer to procedure for Longitudinal Drive Calibration.

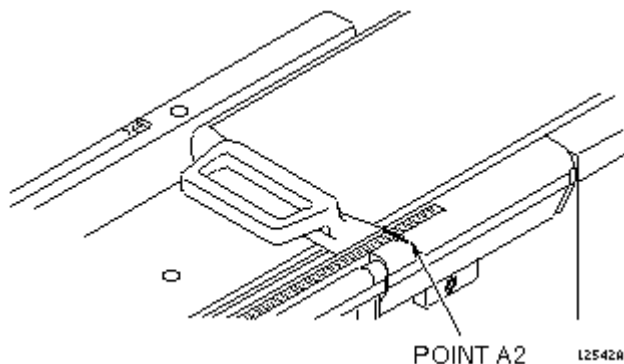
1- LONGITUDINAL DRIVE CALIBRATION

1. During Load From Cold, *Begin of cradle travel* should have been set to 50 and *End of travel* set to 2550. This procedure assumes this starting point. If values are different, change them back to 50 and 2550, and download the TPS/ISE before continuing, procedure for Configuration File.
2. Move the cradle to the home position.
3. Move the cradle approximately 10 cm into the bore.
4. Turn on the alignment lights and press landmark to set the position display on the gantry to zero.
5. Place a piece of tape on the cradle near the point farthest from the magnet and another piece aligned with it on the edge of the table. See Illustration L2541A.



MARKING/INDEXING REFERENCE POINT A1
ILLUSTRATION L2541A

6. Fast forward the cradle into the bore until the first piece of tape is located near the end of the table.
7. Place a third piece of tape on the edge of the table aligned with the first piece that is on the cradle. There are now two pieces of tape on the table. The two pieces of tape are spaced apart the same distance that the cradle has moved. See Illustration L2542A.



READING POINT A2
ILLUSTRATION L2542A

8. Move the cradle position displayed on the gantry. This is the *displayed distance*.
9. Measure the distance on the table between the second and third pieces of tape. This is the *measured distance*. Move the table away from the magnet if necessary to use a steel rule but DO NOT move the cradle. The display on the gantry should not change, (1" = 25.4 mm).
10. The new *End of cradle travel* variable = $[2500 \text{ (displayed distance)} / \text{(measured distance)}] + 50$.
11. Enter this new value in the configuration file.
12. Download the TPS/ISE.

13. After the download, the cradle travel distance displayed on the gantry will equal the measured distance. Illustration L2542A shows table, cradle 10 cm from home position, bore of magnet, and the first two pieces of tape. Illustration L2541A shows same as L2542A, but with end of cradle near the bore of the magnet and the addition of the third piece of tape.

1-1 Tools Required

1. A nonferrous scale that can accurately measure 800mm or more.

WARNING!

POSSIBLE PERSONAL INJURY AND/OR EQUIPMENT DAMAGE! TOOLS MADE OF FERROUS OR MAGNETIC MATERIAL MAY BECOME DANGEROUS PROJECTILES AND CAUSE EQUIPMENT DAMAGE OR BODILY INJURY WHEN USED NEAR THE MAGNET. USE ONLY A NONFERROUS SCALE FOR THIS PROCEDURE.

Note

Longitudinal Drive Calibration must be performed before the Electrical Isocenter Calibration procedure.

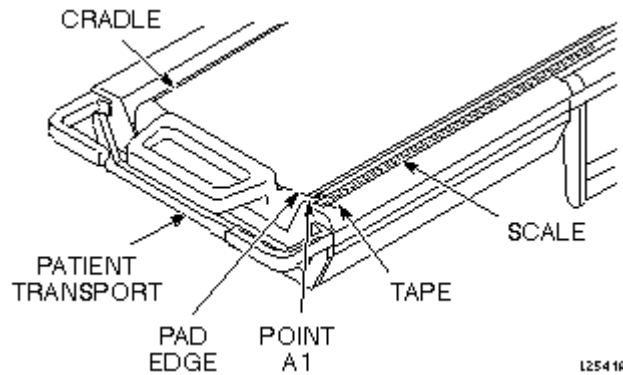
1-2 Setting 2500 mm Beginning and End of Travel Parameters

1. Locate service desktop manager and click on **[C Shell]** .
 2. Edit the config file by typing **cfe<Enter>**.
- (Refer to procedure for Configuration File, Section 1, for addition information on file editing commands if needed.)
3. Search file for 2500 mm travel parameters by scrolling down with the **<Enter>** button.
 4. Change the parameter beg Travel to 50 mm.
 5. Change the parameter end Travel to 2500 mm.
 6. Save the changes just made and quit the configuration file editor by typing **<Ctrl>Q**.
 7. Close the configuration file editor by typing **<Ctrl>Q** again.
 8. **[SYSTEM SHUTDOWN]** to bring down the system.

1-3 Determine End Of Travel Calibration Constant

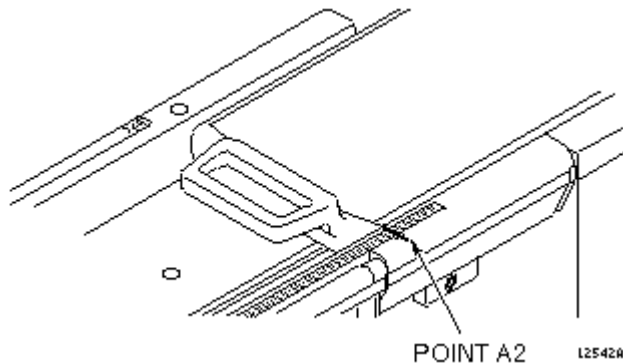
1. **[Reset TPS]**

2. After system has initialized, move the cradle into magnet until longitudinal display reads between 50 to 100 (mm); use IN FAST and IN SLOW buttons on either control switch board.
3. Apply a piece of tape, aligned with the carriage pad edge, to the patient transport. See Illustration L2541A.



MARKING/INDEXING REFERENCE POINT A1
ILLUSTRATION L2541A

4. Place the scale along the patient transport and align a convenient reference point at the end of the scale to the tape edge (e.g., 10, or 800 mm). Record the scale point as A1 in Table 1. Record the shroud display reading as D1.
5. Move the table to a point near the opposite end of the scale, at least 700 mm from the first point, using the IN FAST and IN SLOW buttons. See Illustration L2542A.



READING POINT A2
ILLUSTRATION L2542A

Note

To eliminate backlash errors, be sure to approach the point from the same direction from which the first point was approached.

6. Record the scale reading as A2 and the position on the shroud display as D2 in Table 1.
7. Calculate the actual distance travelled as $DA = |(A2-A1)|$, and record value in Table 1.
8. Calculate the shroud distance as $DD = |(D2-D1)|$, and record value in Table 1. If DD differs from DA by 1 mm or less, no further calibration is needed. If DD differs from DA by more than 1 mm, continue with this procedure.
9. Calculate the parameter End of 2500 mm Travel in Table 1 (round off calculated value to nearest mm).

1-4 Update Configuration File

1. Edit the configuration file and using the instructions in Section 1-1 change the parameter end Travel to the calculated End of Travel value from Table 1 2500 mm.

Table 1

A1= D1=

A2= D2=

DD (D2-D1)=

DA (A2-A1)=

End of 2500 mm Travel $(2500 * DD/DA)+50=$

Note

Calculated End of 2500mm Travel is typically around 2570mm and cannot exceed 2600mm.

1-5 Check Calibration

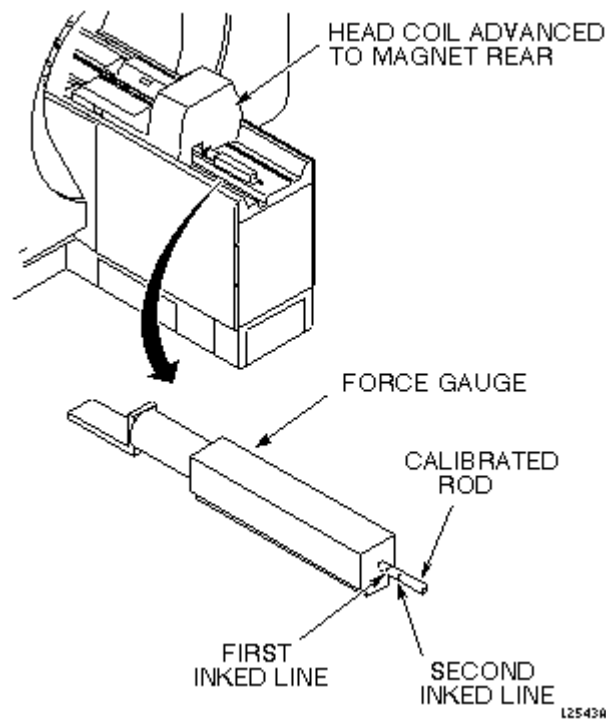
1. Repeat Section 1-2, Setting 2500mm Beginning and End of Travel Parameters.
2. DD should now equal DA within 1 mm; if it does not, repeat Section 1-2 and Section 1-3 and recheck calibration.

2- LONGITUDINAL DRIVE CLUTCH TENSION ADJUSTMENT

This procedure uses the Longitudinal Drive Force Gauge (46-307500G1) to adjust Longitudinal Drive Motor (MG3 A7) clutch tension on the Longitudinal Drive Belt (46-282388P1).

1. Position head coil carriage to a location inside magnet bore.

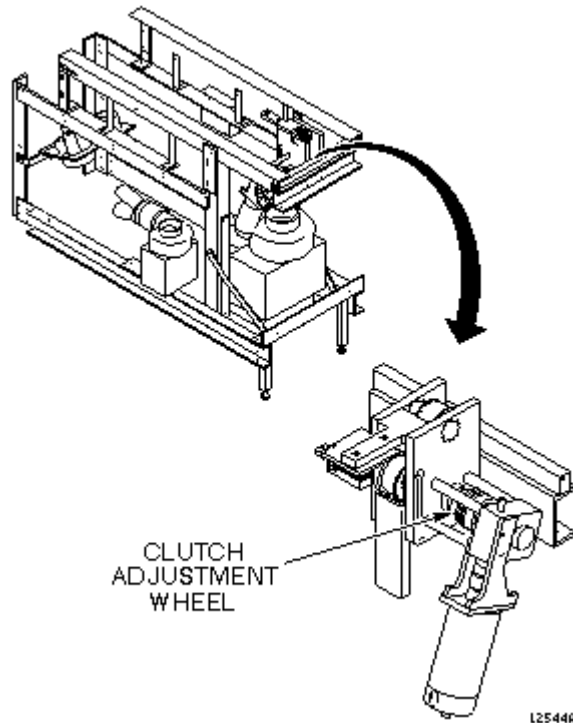
2. Position force gauge on top of longitudinal drive belt at very end of belt track, magnet rear (see Illustration L2543A).



CARRIAGE CLUTCH TENSION
ILLUSTRATION L2543A

3. Push calibrated rod of force gauge all the way into body of force gauge (fully retracted).
4. Press IN SLOW button on magnet enclosure until head coil comes to a stop against force gauge. The force of the advancing head coil causes the force gauge calibrated rod to protrude from rear of force gauge.
5. Inspect inked calibration lines on calibrated rod. If first inked line is not visible (clutch tension is too small), or if portion of rod behind second inked line is visible (clutch tension is too great), clutch tension requires adjustment; continue with procedure. If rod protrudes such that only the portion between the two inked lines is visible, no further adjustment is needed.

6. Remove rear pedestal cover to gain access to clutch tension adjustment wheel. See Illustration L2544A.



LONG DRIVE CLUTCH ADJUST WHEEL
ILLUSTRATION L2544A

7. Turn clutch adjustment wheel clockwise to tighten (increase tension) or counterclockwise to loosen (decrease tension).
8. Repeat steps 1 through 5 to determine if further calibration is required.
9. Replace cover on rear pedestal.

3- LONGITUDINAL DRIVE MOTOR ALIGNMENT TO FIELD

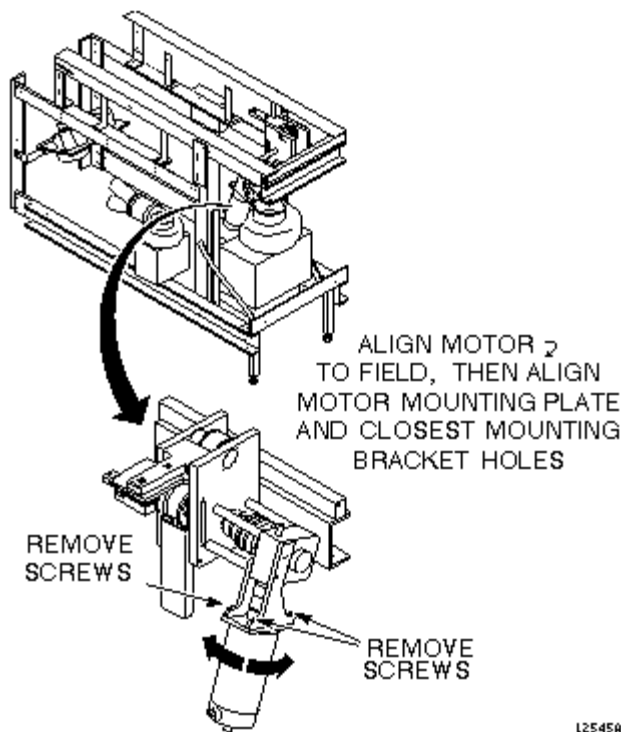
This procedure must be performed on either a replacement motor, or upon initial system installation to insure that the motor is properly aligned to the magnet field.



FERROUS MATERIAL HAZARD!! THE MOTOR IS MADE OF FERROUS MATERIAL. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE, OR MAGNET, DO NOT LIFT THIS MOTOR ABOVE THE LEVEL OF THE MOUNTING ON THE REAR PEDESTAL FRAME. TWO PERSONS ARE REQUIRED TO REPLACE THE MOTOR.

1. Remove power to Long Drive Motor by switching off circuit breakers:
RF/PEN Cabinet: Lock out and tag out circuit breaker MR1 A14 CB6 at rear of RF/Pen Cabinet on magnet enclosure power supply module.
RF/PEN II Cabinet: Lock out and tag out circuit breaker MR1 A20 CB4 at rear of RF/Pen II Cabinet.
(Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Verify that voltage between 38.5V Dock and Dock / Light RTN is 0 Vdc.

3. Remove four mounting bracket screws and loosen lower set screw, if initial system installation. See Illustration L2545A.



LONG DRIVE MOTOR ALIGNMENT
ILLUSTRATION L2545A

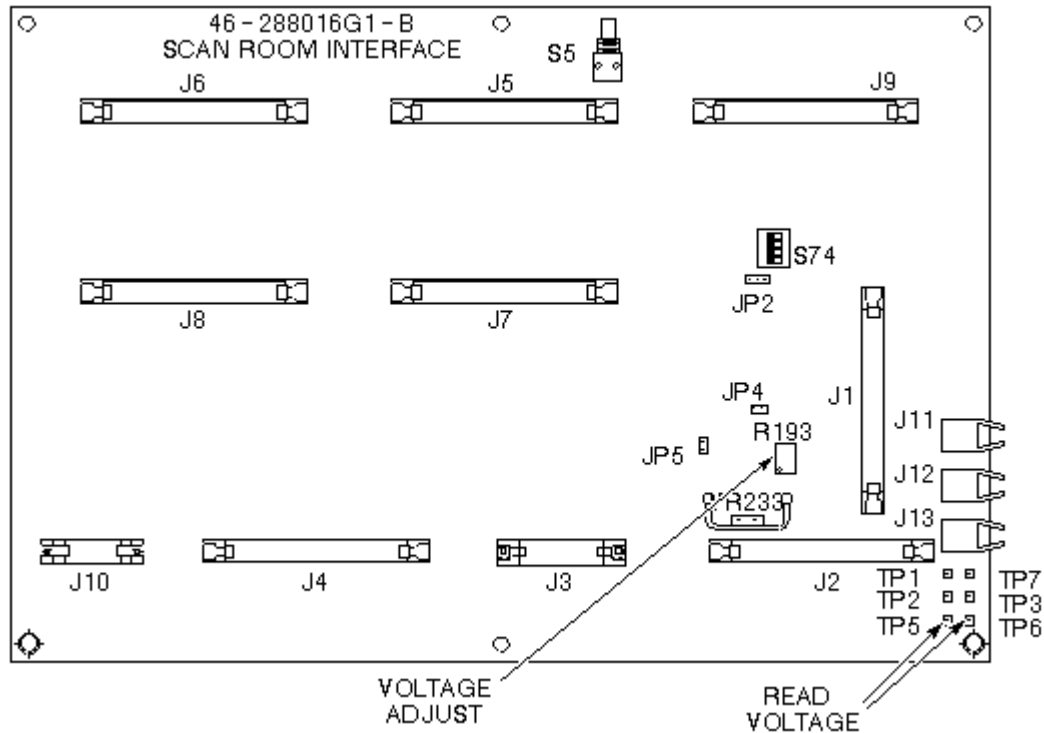
4. Holding motor firmly with both hands, and staying as far away from magnet as possible, loosen two mounting bracket screws and insert motor assembly into shaft coupling on rear pedestal.
5. Carefully loosen grip on motor just enough to allow motor to align itself to magnetic field. See Illustration L2545A.
6. One person holds motor and aligns motor to mounting bracket screw holes, while second person installs four screws to secure motor assembly to mounting bracket.
7. Tighten lower set screw on motor shaft coupling.
8. Connect wires onto terminals of motor: black leads to black screw cap, multicolored leads to red screw cap.
9. Install rear pedestal left side cover.
10. Remove lock out and tag out devices from circuit breaker MR1 A14 CB6 at rear of RF/PEN Cabinet, or MR1 A20 CB4 at the rear of the RF/PEN II Cabinet.
11. Restore power to SRI by switching on circuit breaker MR1 A14 CB6 at rear of RF/PEN Cabinet, or MR1 A20 CB4 at the rear of the RF/PEN II Cabinet.

12. Verify that pressing IN FAST button on Control Switch Board (MG2A4, A5) moves cradle into magnet bore, and that pressing OUT FAST button moves cradle out of magnet bore. If cradle movement is opposite of what it should be, reverse wires on terminals of longitudinal motor.

4- LONGITUDINAL ENCODER SUPPLY VOLTAGE ADJUSTMENT

Adjustment of the five volt power going to the longitudinal encoder (MG3 A6) is performed on the SRI (scan room interface) board.

1. Remove right front magnet enclosure cover to gain access to the SRI module (MG2 A33).
2. Loosen SRI module cover screws and slide cover to the left. See Illustration L2546A.



LONGITUDINAL ENCODER SUPPLY VOLTAGE ADJUSTMENT
ILLUSTRATION L2546A

3. With system powered up, connect meter set for dc volts to TP5 and TP6 (see Illustration L2546A).
4. Adjust R193 (located above J2 and left of J1) until a 5.0 ± 0.1 Vdc reading is indicated on the meter.
5. Slide SRI module cover closed and tighten cover screws.
6. Install magnet enclosure cover.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	April 17, 1998	W. Pasciak	Initial version released converted to Word
0	May 7, 1998	wek	Updated lock out/tag out info
1	Oct 13, 1999	M. Keber	Added missing title; heading still had Rev A.